
MORGAN PFAFF

❖ morgan.pfaff1@gmail.com ❖ [linkedin.com/in/morgan-pfaff](https://www.linkedin.com/in/morgan-pfaff) ❖ morganpfaff.github.io ❖

EDUCATION

University of California (UC) Berkeley and UC San Francisco

Berkeley, CA

Doctor of Philosophy in Bioengineering

December 2026

- GPA 4.0/4.0
- National Science Foundation Graduate Research Fellowship Awardee
- Major area of study: Biomaterials Engineering and Tissue Regeneration
- Minor area of study: Cell, Tissue, and Developmental Biology
- Completed rotations in the laboratories of Dr. Kevin Healy, Dr. Phillip Messersmith, and Dr. Sanjay Kumar

Northeastern University

Boston, MA

Bachelor of Science in Biochemistry

May 2021

- GPA 4.0/4.0
- Honors Program Member and Honors in the Discipline
- Academic courses include advanced material such as Protein Chemistry, Cellular & Molecular Biology, Biochemistry, Biostatistics, Organic Chemistry, and Physical Chemistry, as well as a general survey of Computer Science, Physics, and Calculus
- Participated in an Engineering study abroad program in Oxford, United Kingdom in Summer 2019

RESEARCH EXPERIENCE

MuscleMatrix

Berkeley, CA

Scientist

August 2026 – Present

- Engineering hyaluronic acid-based hydrogels for volumetric muscle loss

Kevin Healy Lab, UC Berkeley

Berkeley, CA

PhD Candidate

January 2022 – August 2026

- Engineered tunable, semi-synthetic hyaluronic acid (HyA) hydrogels with viscoelastic properties designed to promote muscle regeneration and vascularization after complex soft tissue injuries, including delayed rotator cuff repair and volumetric muscle loss
- Investigated hydrogel-driven mechanisms of muscle regeneration, including induction of brown adipose tissue (BAT) differentiation in fibro-adipogenic progenitors (FAPs), using 3D *in vitro* systems
- Co-inventor on MuscleMatrix, translating hydrogel formulations from bench-scale characterization toward a therapeutic device with near-term clinical potential
- Communicate findings through 2 peer-reviewed publications and 4 national and international conference presentations (BSRM, BMES, SFB, ORS), collaborating across orthopedic surgery, biomaterials, and cell biology teams

Directed Assembly of Particles and Suspensions, Northeastern University

Boston, MA

Biomaterials Undergraduate Researcher

September 2016 – July

2021

- Wrote a grant proposal for and received the prestigious Project-Based Exploration for the Advancement of Knowledge Summit Award to fund my research
- Developed a protocol to fabricate dual-network hydrogels with magnetically-responsive hydroxyapatite fiber reinforcement for bone graft applications and tailored composite design to maximize toughness
- Characterized and investigated the mechanical strength of hydrogel samples by independently conducting sensitive experiments using scanning electron microscopy and dynamic mechanical analysis
- Designed, ran, and analyzed simulations using COMSOL Multiphysics FAE software of hydroxyapatite fiber rotation in an alginate matrix to investigate a potential new toughening mechanism

- Manufactured freeze-casted conduits with highly aligned, double layered porosity for peripheral nerve repair in collaboration with Dartmouth College and Harvard Medical School

Vertex Pharmaceuticals

Boston, MA

Exploratory Research Co-op
2020

January 2020 – June

- Developed and optimized cellular assays for transcription factor detection in potential druggable pathways for the treatment of Sickle Cell Disease (SCD)
- Conducted pilot experiments to determine expression profiles of various proteins over time
- Utilized molecular cloning of recombinant vectors for stable integration into cells line of interest
- Independently planned, executed, and analyzed results from experiments and regularly communicated findings with other members in the Exploratory Research group
- Analyzed and presented on relevant articles during Journal Clubs in front of the SCD team

Editas Medicine

Cambridge, MA

Screening Group Co-op
2018

January 2018 – June

- Screened for the most effective sgRNAs for the company's pipeline diseases and performed CRISPR-mediated mutations of genes of interest via nucleofection, transfection, and viral transduction
- Programmed and optimized high-throughput screening assays for automated liquid handlers
- Conducted a complete comparative analysis of the company's Biomek i7 automated liquid handler and its Biomek FX predecessor
- Utilized various computer software such as Prism, Tableau, and FlowJo to visualize, analyze, and compare data and presented findings to other members in the screening group and other departments

University of Connecticut Health Center

Farmington, CT

Summer Research Fellow
2018

June 2017 – August

- Piloted synthetic-lethal experiments to find exploitable vulnerabilities in cell signaling pathways and elucidate novel druggable osteosarcoma targets
- Tested the effects of various inhibitor and siRNA combinations on undifferentiated hFOB cells and osteosarcoma cell lines SaOS2 and U2OS to specifically target tumor cells with mutations interfering with growth factor receptor regulation
- Practiced sterile technique to grow and maintain cell cultures for several months and completed analyses using PCR and absorbance assays
- Work resulted in a poster presentation and provides the foundation for NIH grants from this lab

RESEARCH SKILLS

Cellular & Molecular Biology

- Aseptic Technique, Automated Liquid Handling (Biomek i7, Biomek FX, Mantis, Tempest), ddPCR, DNA Extraction, DNA Purification, Flow Cytometry, Gel Electrophoresis, High-Powered Cell Disrupt, Laser Cell Capture, Lipofection, Mammalian Cell Culture, Molecular Cloning, Nucleofection, qPCR, Restriction Digest, Viral Transduction, Western Blot

Material Fabrication/Characterization

- Dynamic Mechanical Analysis, Freeze Casting, Lyophilization, Nuclear Magnetic Resonance (NMR) spectroscopy, Optical Microscopy, Scanning Electron Microscopy, Sonication, Tensile Testing

Programming

- Proficient in LaTeX, Java, Microsoft Office (Word, Excel, PowerPoint, Outlook, etc.)
- Familiar with C++, MATLAB, Arduino

Statistical Analysis, Graphing, and Simulation Software

- R, Tableau, GraphPad Prism, FlowJo, SnapGene, COMSOL Multiphysics

ADDITIONAL WORK EXPERIENCE

Tufts Medical Center, Oncology and Hematology

Boston, MA

Clinical Care Technician
2019

January 2019 – December

- Assisted 6-10 patients per shift with tasks of living such as ambulation, bathing, dressing, and eating
- Monitored patient vital signs (blood pressure, temperature, pulse, O₂ saturation), blood sugar levels, and overall patient condition and reported abnormal conditions to the assigned nurse
- Supported nurses and physicians during examinations, medical procedures, and emergency situations
- Responded to the needs of patients and family members and took a proactive role in the efficient operation of the hospital

Northeastern University Housing and Residential Life

Boston, MA

Resident Assistant
2019

August 2017 – July

- Served as a live-in mentor and resident advisor for ~45 freshman women a year, maintaining a safe, orderly environment and confronting and resolving issues surrounding diversity, racism, substance abuse, academic performance, and social acceptance
- Designed and implemented educational, recreational, and networking programs, including Q&A panels with graduate students and professors, leadership workshops, and faculty mentorship events
- Helped residents set and achieve short- and long-term goals, connect with established scientists and professors, and access resources to pursue their academic and scientific aspirations

Fallon Ambulance Service

Quincy, MA

Emergency Medical Technician Basic
2019

September 2018 – April

- Conducted medical and psychological assessments of ill or injured patients, provided them with the appropriate immediate care, and safely transported them to the appropriate hospital or specialized medical facility
- Interacted with patients and families during transports and emergencies and maintained a calm, safe environment for all parties
- Collected pertinent information including vitals, symptoms, medical history, allergies, and prescriptions from patients and recorded information according to company protocols in the appropriate database

Dental Office of Dr. Andrew J. Ponichtera DMD, MS

Simsbury, CT

Dental Assistant
2016

June 2016 – August

- Assisted the dentist with procedures including fillings, root canals, and insertion of crowns, implants, bridges, and dentures
- Handled front desk operations including greeting patients, booking and confirming appointments, and organizing patient files
- Took inventory of dental supplies and equipment and placed new orders as needed

PUBLICATIONS & PRESENTATIONS

Publications

1. Wang, V. Y., Zhu, Y., Spaulding, K. A., Pfaff, M., Neiman, G., Takaba, K., Chen, N., Parekh, K., Hasaballa, A., Haraldsson, H., Baker, A., Saloner, D., Wallace, A. W., Ziats, N. P., Healy, K. E., Ratcliffe, M. B. NIPAAm-co-PEG4000 thermoresponsive hydrogel with ROS scavenging properties improves in vivo border zone contractility and reduces myocardial remodeling in sheep after MI. *Acta Biomaterialia*, 2026, 211, 48–60. doi.org/10.1016/j.actbio.2025.06.046
2. Pfaff, M. R., Wague, A., Davies, M., Killaars, A. R., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Viscoelastic HyA hydrogel promotes recovery of muscle quality and

vascularization in a murine model of delayed rotator cuff repair. *Advanced Healthcare Materials*, 2025, 14, 2403962. doi.org/10.1002/adhm.202403962

Oral Presentations

1. **Pfaff, M. R.**, Wague, A., Davies, M., Killaars, A. R., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. Oral presentation presented at: Biological Scaffolds for Regenerative Medicine (BSRM); May 2025; Napa, CA.
2. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. E. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Oral presentation presented at: Biomedical Engineering Society (BMES) Annual Meeting; October 11–14, 2023; Seattle, WA.
3. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Oral presentation presented at: Society For Biomaterials Annual Meeting & Exposition, "Biomaterials for Regenerative Engineering"; April 21, 2023; San Diego, CA.
4. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Oral presentation presented at: Northeastern Regional Student Conference of the Society for Experimental Mechanics; June 22, 2020; Dartmouth, MA.

Poster Presentations

1. **Pfaff, M.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Poster session presented at: ORS 50th International Musculoskeletal Biology Workshop; July 22–27, 2023; Midway, UT.
2. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Poster session presented at: Society For Biomaterials (SFB) Annual Meeting & Exposition; April 2023; San Diego, CA.
3. **Pfaff, M.**, Erb, R. Understanding rod rotation in fiber reinforced hydrogels: a potential toughening mechanism. Poster session presented at: Research, Innovation, and Scholarship Expo; April 15, 2021; Boston, MA.
4. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Poster session presented at: Materials Research Society 2020 Spring/Fall Meeting and Exhibit; November 28, 2020; Boston, MA.
5. **Pfaff, M.**, Alander, C., Hansen, M. Synthetic lethal approaches to treating osteosarcoma. Poster session presented at: University of Connecticut Health Center Summer Research Fellowship Program; July 27, 2017; Farmington, CT.
6. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Poster session presented at: Research, Innovation, and Scholarship Expo; April 13, 2017; Boston, MA.
7. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Poster session presented at: National Conference on Undergraduate Research; April 6–8, 2017; Memphis, TN.

FELLOWSHIPS, AWARDS, AND HONORS

Fellowships, Grants, and Scholarships

- National Science Foundation Graduate Research Fellowship 2021
- Fellowship awarded to outstanding graduate students who are pursuing research-based master's and doctoral degrees in fields within NSF's mission (\$138,000)
- Project-Based Exploration for the Advancement of Knowledge "Shout It Out!" Award 2020
- Conference and travel grant awarded to undergraduate researchers who have been accepted to present the outcome of their project at an external venue or conference

- Project-Based Exploration for the Advancement of Knowledge "Summit" Award 2020

- Research grant awarded to undergraduate researchers with the skills and sophistication to work independently, the ability to formulate creative research questions, and the experience to develop and pursue pathways to work through them (\$3,000)

Northeastern University Conference Travel Fund Award 2017

- Travel grant awarded to students presenting their original research at professional conferences

Presidential Global Scholarship 2016

- Scholarship awarded to outstanding students who show commitment to academic success and interest in pursuing global experiences (\$6,000)

Northeastern University Connections Merit Scholarship 2016

- Merit scholarship awarded to female students entering STEM in the top 10-15% of the application pool based on SAT score, high school grades, and class ranking (\$18,000/year)

Awards and Honors

RISE Excellence in Research Award 2021

- Awarded to one out of over 450 presenters for methodological and procedural excellence in exploration, creation, and study at the Northeastern University Research, Innovation, and Scholarship Expo (\$2,000)

Northeastern University Huntington 100 2021

- Awarded to 100 outstanding students each year for achievements which are commensurate with the university's mission, ideals, values, and Academic Plan

Sigma Xi Scientific Research Honor Society 2020-2021

- Membership awarded to students who have conducted independent investigation resulting in new scientific discoveries and produced written reports concerning their research

Northeastern University President's Award 2020-2021

- Awarded to the top 10 students of each university class as measured by GPA

Northeastern University Dean's List 2016-2021

- Awarded to the top 10-15% of students in the College of Science each semester

RISE Innovation Alley 2017

- Poster presentation selected to be in the main corridor of projects with the highest impact and commercial potential at the Northeastern University Research, Innovation, and Scholarship Expo

COMMUNITY OUTREACH

Food Not Bombs September 2024 –

Present

- Coordinate weekly food distribution with Food Not Bombs East Bay, picking up vegetarian meals from the cookhouse, occasionally helping package meals into boxes, and transporting and setting up a serving table at a designated intersection
- Serve 25-40 free vegetarian meals per shift over roughly 2 hours to unhoused and low-income community members
- Build ongoing relationships with unhoused and low-income residents, fellow volunteers, and community members interested in getting involved, deepening a personal commitment to food justice and equitable access

Adventure Challenge Volunteer September 2018 – December

2019

- Planned activities for and accompanied my Adventure Challenge partner, a 13-year old girl with autism, to the YMCA pool to swim and played pool games on a weekly basis
- Promoted physical activity, social interaction, and overall health and wellness to help her maintain a healthy lifestyle

Resident Assistant
2019

August 2017 – July

- Served as a mentor for 45 freshman women from around the world looking to pursue a career in STEM
- Helped my residents set and achieve short- and long-term goals, network with established scientists and professors, and access resources to fulfill their academic and scientific aspirations
- Designed programs including Q&A panels with graduate students and professors, leadership and team building workshops, and networking events with professors looking to mentor new students

Advancement Via Individual Determination Tutor
2018

September 2018 – December

- Served as a discussion mediator during highly interactive tutoring sessions for students participating in the AVID program
- Assisted students in learning how to guide each other towards solutions to various problems via questions and critical thinking in order to develop intellectually and acquire effective, long-term study habits
- Promoted skills that would allow students to more feasibly pursue and succeed in higher education after graduating high school

University of Connecticut Health Center Volunteer
2018

June 2018 – August

- Greeted and seated incoming patients arriving for injections or surgery at the Musculoskeletal Institute at UCONN Health Center
- Alleviated pressure on front desk staff by assisting with front desk operations including assembling patient charts, pulling and organizing patient files, and answering phone calls

ORGANIZATIONS & GROUPS

Sigma Xi, The Scientific Research Honor Society

March 2020 – Present

- 2021-Present: Berkeley Chapter Vice President; Chairperson of the Committee for Student Advancement, a committee I developed to enhance student representation and success in the Berkeley Chapter
- 2020-2021: Diversity & Inclusion Chair and Associate Member of the Northeastern University Chapter

Northeastern University Club Sports Council
2021

September 2019 – June

- Served as a liaison between Club Sports teams and Club Sports Administrators and Directors

Northeastern University Triathlon Team
2021

September 2016 – June

- President/Captain (2019-2020), Chief Financial Officer (2018-2019), Chief Procurement Officer (2017-2018)

CERTIFICATIONS

U.S. Center for SafeSport Certification

2019

- SafeSport certified for completion of training on mandatory reporting, sexual misconduct awareness education, and emotional and physical misconduct among youth and student athletes

Emergency Medical Technician, Basic

2018

- Certified EMT-B in Massachusetts, Connecticut, and Nationally